AYE
TECH HUB

• MECHANICAL ENGINEERING • LESSON 05

# Fluid Mechanics

Pressure · Bernoulli · Pipe Flow · Reynolds · Pumps

Lesson 05 of 30 | Mechanical Engineering Program


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- Pressure, density and viscosity — the three core fluid properties
- Bernoulli's equation — the most important equation in fluid mechanics
- Laminar vs turbulent flow and the Reynolds number
- Pipe flow: head loss, Darcy-Weisbach and the Moody chart
- Pump types, affinity laws and pump selection for engineers

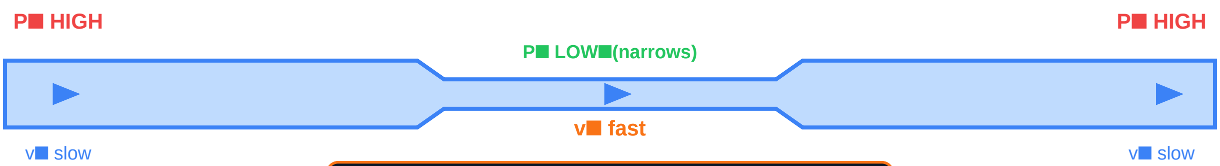
# Fluid Mechanics — The Science of Flowing Matter

Fluid mechanics is the branch of physics that studies the behaviour of **liquids and gases at rest (fluid statics) and in motion (fluid dynamics)**. For mechanical engineers, it is the foundation of pump design, pipe system analysis, aerodynamics, hydraulics, HVAC duct design, and turbomachinery. Every system that moves fluid — oil, water, air, gas — is governed by fluid mechanics principles.

## Key Fluid Properties

| Property | Symbol | Definition | Unit | Engineering Impact |
|---|---|---|---|---|
| Density | ρ (rho) | Mass per unit volume. Water: 1000 $kg/m^3$. Air: 1.2 $kg/m^3$. | $kg/m^3$ | Determines buoyancy, pressure, inertial forces in flow |
| Viscosity (dynamic) | μ (mu) | Resistance to flow. "Thickness" of fluid. Water: 0.001 Pa·s. Oil: 0.03–3 Pa·s. | Pa·s ($N{\cdot}s/m^2$) | Governs pipe friction, bearing lubrication, pump selection |
| Viscosity (kinematic) | $\nu = \mu/\rho$ | Dynamic viscosity ÷ density. Used in Reynolds number. | $m^2/s$ | Used in all dimensionless flow calculations |
| Pressure | P | Normal force per unit area. Atmospheric: 101 325 Pa = 1 atm = 1.013 bar. | Pa ($N/m^2$) | Drives flow; determines pipe wall thickness and pump head |
| Surface tension | σ (sigma) | Cohesive force at liquid surface. Causes capillary action. | N/m | Relevant in small-scale systems, droplet formation, microfluidics |

## Bernoulli's Equation — The Most Important Equation in Fluid Mechanics




$$P + \tfrac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

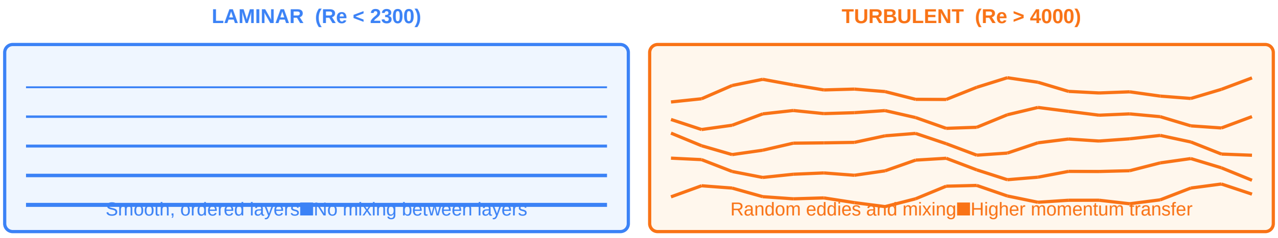
*Figure 1 — Bernoulli's Equation: Venturi Effect — Pressure Drops Where Velocity Rises*

| Term | Name | Physical Meaning | Units |
|---|---|---|---|
| P | Static pressure | Pressure of the fluid at rest — the "push" energy per unit volume | Pa ($N/m^2$) |
| $\tfrac{1}{2}\rho v^2$ | Dynamic pressure | Kinetic energy per unit volume — energy of motion | Pa |
| $\rho g h$ | Hydrostatic pressure | Potential energy per unit volume — energy of elevation | Pa |
| $P + \tfrac{1}{2}\rho v^2 + \rho g h$ | Total (stagnation) pressure | Sum is constant along a streamline in steady, inviscid flow | Pa |

***Key insight from Bernoulli: When a fluid speeds up (velocity ↑), its pressure must drop (P ↓) to keep the sum constant. This is why aircraft wings generate lift, why carburettors atomise fuel, why a venturi meter measures flow rate, and why shower curtains blow inward when water flows.***

# Reynolds Number and Pipe Flow

The **Reynolds number (Re)** is the single most important dimensionless number in fluid mechanics. It predicts whether flow in a pipe or over a surface will be smooth and orderly (laminar) or chaotic and mixing (turbulent). Almost every engineering flow calculation depends on knowing whether Re is above or below the critical threshold.




Transition zone: 2300 < Re < 4000

*Figure 2 — Reynolds Number: Laminar vs Turbulent Flow*

| Flow Regime | Reynolds Number | Characteristics | Practical Examples |
|---|---|---|---|
| Laminar | Re < 2 300 | Smooth, parallel layers. Parabolic velocity profile. Low friction. Very predictable. | Oil in small pipes; blood flow in capillaries; slow water in large pipes |
| Transitional | 2 300 < Re < 4 000 | Unstable — switches between laminar and turbulent. Avoid in engineering design if possible. | Design systems to operate clearly in one regime |
| Turbulent | Re > 4 000 | Chaotic eddies. Flat velocity profile. Higher friction. Better heat and mass transfer. | Most industrial pipe flow; air in ducts; water supply systems |

## Reynolds Number Formula

| Formula | Variables | Example |
|---|---|---|
| $Re = \rho VD / \mu = VD / \nu$ | ρ = density (kg/m³) V = mean velocity (m/s) D = pipe diameter (m) μ = dynamic viscosity (Pa·s) ν = kinematic viscosity (m²/s) | Water (20°C) in 100mm pipe at 2 m/s: Re = 2 × 0.1 / (1×10■■) = 200 000 → Turbulent ✓ |

## Pipe Flow — Head Loss and the Darcy-Weisbach Equation

| Equation / Concept | Formula | Notes |
|---|---|---|
| Darcy-Weisbach head loss | $h_f = f \times (L/D) \times (V^2/2g)$ | f = Darcy friction factor (from Moody chart); L = pipe length; D = diameter; V = velocity |
| Friction factor (turbulent) | Use Moody chart or Colebrook equation: $1/\sqrt{f} = -2 \log(\varepsilon/3.7D + 2.51/Re\sqrt{f})$ | ε = pipe roughness (mm): steel=0.046, cast iron=0.26, concrete=0.3-3 |
| Friction factor (laminar) | $f = 64 / Re$ | Only valid for Re < 2300. Simple and exact. |
| Minor losses (fittings, bends) | $h_m = K \times (V^2/2g)$ | K = loss coefficient: elbow 0.5–1.5; valve fully open 0.1–0.5; entry 0.5 |
| Total head loss | $h_{total} = h_f + \Sigma h_m$ | Sum of friction losses and all fitting losses along the pipe system |
| Continuity equation | Q = A■V■ = A■V■ (ρ■A■V■ = ρ■A■V■ for compressible) | Mass (or volume for incompressible) is conserved — flow rate is constant |

# Pumps — Types, Selection and the Affinity Laws

A pump converts mechanical energy (from a motor) into hydraulic energy (pressure and flow) in a fluid system. Selecting the right pump type and size is one of the most common mechanical engineering tasks. It requires understanding the system's **required flow rate, total head, and the fluid's properties**.

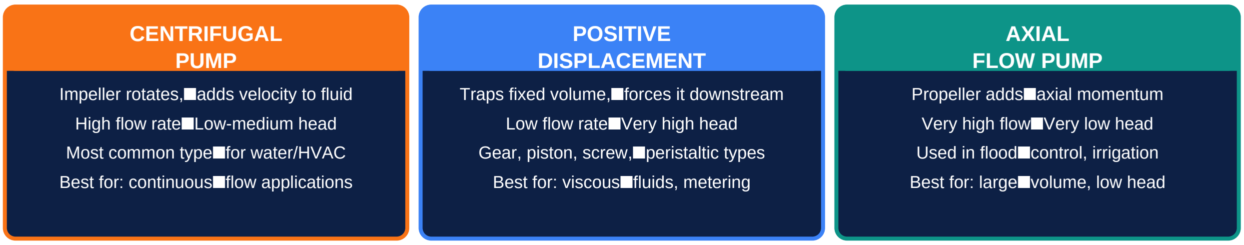



*Figure 3 — Pump Types: Centrifugal / Positive Displacement / Axial*

## Pump Selection — The System Curve and Pump Curve

| Concept | Definition | Engineering Application |
|---|---|---|
| Head (H) | Energy added to fluid per unit weight. Total head = static head + friction head + velocity head. | H (metres) = pump must overcome elevation + pipe losses + entry/exit losses |
| Flow rate (Q) | Volume of fluid delivered per unit time ($m^3/s$, L/s, or $m^3/h$) | Determined by system demand — number of outlets, required velocities |
| System curve | Plot of required head vs flow rate for the pipe system. $H_{sys}$ = static head + k × $Q^2$ | Parabolic — more flow requires more head due to increased pipe friction |
| Pump curve | Manufacturer's plot of pump head vs flow rate. Head decreases as flow increases. | Operating point = intersection of pump curve and system curve |
| Efficiency curve | Best Efficiency Point (BEP) is the peak of the pump efficiency curve. | Always select pump so operating point is within ±10% of BEP |
| NPSH (Net Positive Suction Head) | Minimum pressure needed at pump inlet to prevent cavitation. | $NPSH_{available} > NPSH_{required}$ — check before finalising pump selection |

## Pump Affinity Laws

| When speed changes from N■ to N■: | Formula | Practical Example |
|---|---|---|
| Flow rate scales with speed | Q■ = Q■ × (N■/N■) | Doubling speed doubles flow |
| Head scales with speed squared | H■ = H■ × (N■/N■)² | Doubling speed quadruples head |
| Power scales with speed cubed | P■ = P■ × (N■/N■)³ | Doubling speed → 8× the power! This is why VFDs save massive energy at part load |

***Variable Frequency Drive (VFD) energy saving: Running a pump at 80% speed instead of 100% uses only $0.8^3$ = 51% of the power — saving 49% of energy. For large pumps running at part load, a VFD pays back its cost in 12–24 months through electricity savings.***

***What comes next — MECH-006: Heat Transfer — the three heat transfer modes (conduction, convection, radiation), the overall heat transfer coefficient, heat exchangers, and thermal resistance networks for engineering design.***